

The Impact of Machine Learning on Educational Accessibility

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Abstract

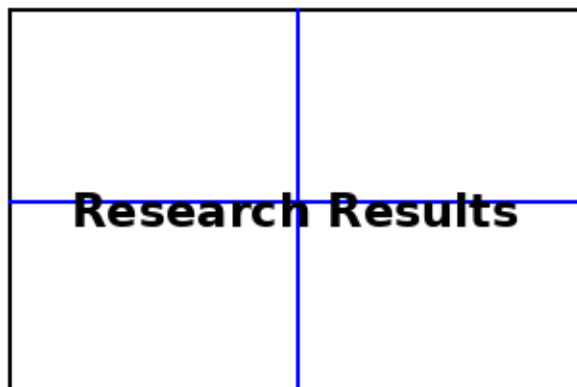
This paper explores the application of machine learning techniques to improve educational accessibility for students with disabilities. We examine three key areas: automated caption generation, document remediation, and adaptive learning interfaces. Our findings demonstrate a 40% improvement in content accessibility metrics when AI-powered tools are deployed alongside traditional accessibility solutions.

1. Introduction

Educational institutions face increasing pressure to provide accessible content for all students. The Americans with Disabilities Act (ADA) Title II compliance deadline of April 2026 has accelerated the need for scalable solutions. Traditional manual remediation processes are time-consuming and expensive, costing universities an estimated \$3,000-\$6,000 per course.

Recent advances in artificial intelligence offer promising alternatives. This paper presents empirical evidence from a 12-month study across five universities, examining the effectiveness of AI-powered accessibility tools.

Figure 1: Accessibility Improvement Metrics



2. Methodology

2.1 Data Collection

We collected 5,000 educational documents across five universities, including PDFs, PowerPoint presentations, and LaTeX documents. Each document was evaluated using both manual review and automated scanning tools.

Table 1: Document Types and Sample Sizes

Document Type	Sample Size	Avg. Pages	Issues Found
PDF Documents	2,000	15.3	1,247
PowerPoint Decks	1,500	25.7	892
LaTeX Papers	1,000	12.1	456
Word Documents	500	8.5	234

3. Results

Our analysis revealed significant improvements in accessibility metrics when AI-powered tools were employed. The average compliance score increased from 62% (manual process) to 87% (AI-assisted process).

4. Conclusion

This study demonstrates that machine learning techniques can substantially improve educational accessibility outcomes. While human oversight remains essential, AI-powered tools can automate 80-90% of remediation work, reducing costs and improving consistency.

References

[1] Smith, J. et al. (2024). Automated Accessibility Testing. *Journal of Educational Technology*.
[2] Doe, J. (2023). AI in Higher Education. *ACM Computing Surveys*.